MATTHEW MILES

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EDUCATION

Doctor of Astrophysics *Mar 2020 - Jul 2024*
Swinburne University of Technology

Thesis title:
Millisecond pulsar timing and gravitational wave searches with the MeerKAT radio telescope

Using a subset of stars known as millisecond pulsars, I worked to construct a cosmic-scale gravitational wave detector known as the MeerKAT Pulsar Timing Array. Ultimately, this will facilitate another method of gravitational wave detection, as well as providing the only method of detection for nanohertz-frequency gravitational waves.
Supervised by: A. Prof. Ryan Shannon, Prof. Matthew Bailes, & Dr. Daniel Reardon

Bachelor of Aerospace Engineering (Honours) and Science *Mar 2014 - Nov 2019*
Monash University

Majors: Aerospace Engineering, Applied Mathematics, Astrophysics

RESEARCH INTERESTS

- Improving pulsar timing and pulsar timing arrays, including the exploration of novel methods (i.e. profile domain / simultaneous polarisation timing).
- Studies of the nature of the stochastic gravitational wave background
- Black Holes / Neutron Stars / Pulsars and their intrinsic properties. (i.e. constraints on the nuclear equations of state, tests of theories of gravity)
- Modelling of noise sources, both common and uncommon to sources in the GW frequency band
- The impact of model misspecification
- Studies of single sources of GWs
- Population modelling of SMBHBs

FIRST-AUTHOR PUBLICATIONS

5. **Miles, M. T.**, et al., 2026, "JUG: JAX-based Unified pulsar timinG"
   The Astrophysical Journal, accepted
4. **Miles, M. T.**, et al., 2025, "The MeerKAT Pulsar Timing Array: The first search for gravitational waves with the MeerKAT radio telescope"
   Monthly Notices of the Royal Astronomical Society
3. **Miles, M. T.**, et al., 2025, "The MeerKAT Pulsar Timing Array: The 4.5-year data release and the noise and stochastic signals of the millisecond pulsar population"
   Monthly Notices of the Royal Astronomical Society
2. **Miles, M. T.**, et al., 2023, "The MeerKAT Pulsar Timing Array: First Data Release"
   Monthly Notices of the Royal Astronomical Society

1. **Miles, M. T.**, Shannon, R. M., Bailes, M., Reardon, D. J., Buchner, S., Middleton, H., and Spiewak, R., 2022, "Mode changing in J1909−3744: the most precisely timed pulsar"
Monthly Notices of the Royal Astronomical Society

CO-AUTHOR PUBLICATIONS

27. Moursy, et al., **(inc. Miles, M. T.)**, 2026, "Pulsar Timing Array Sensitivity to Anisotropy: Empirical Sensitivity Curves, Scaling Relations, and the Multi-Resolution Pixel Basis"
The Astrophysical Journal, submitted

26. Mishra, et al., **(inc. Miles, M. T.)**, 2026, "Interplanetary scintillation-informed heliospheric modelling for the MeerKAT Pulsar Timing Array 4.5 yr dataset"
Monthly Notices of the Royal Astronomical Society

25. Shannon, et al., **(inc. Miles, M. T.)**, 2026, "The SKAO Pulsar Timing Array"
The Open Journal of Astrophysics

24. Agarwal, et al., **(inc. Miles, M. T.)**, 2026, "The NANOGrav 15 yr Data Set: Impacts of Customized Chromatic Noise Models on Gravitational Wave Analyses"
The Astrophysical Journal, submitted

23. Larsen, et al., **(inc. Miles, M. T.)**, 2026, "The NANOGrav 15 yr Dataset: Customized Chromatic Noise Models"
The Astrophysical Journal

22. Di Marco, K., Guttman, N., **Miles, M. T.**, Zic, A., Shannon, R. M., & Thrane, E., 2026, "A transdimensional sampling framework for pulsar timing noise modelling"
The Astrophysical Journal

21. Grunthal, K., Champion, D. J., Thrane, E., Nathan, R. S., Kramer, M., & **Miles, M. T.**, 2026, "Optimising gravitational-wave sky maps for pulsar timing arrays"
Astronomy & Astrophysics

20. Bhat, et al., **(inc. Miles, M. T.)**, 2026, "A glitch in the millisecond pulsar J0900−3144"
Monthly Notices of the Royal Astronomical Society

19. Kulkarni, et al., **(inc. Miles, M. T.)**, 2025, "Measuring scattering variations in pulsar timing observations: a test of the fidelity of current methods"
Monthly Notices of the Royal Astronomical Society

18. Gitika, et al., **(inc. Miles, M. T.)**, 2025, "Optimising the MeerKAT Pulsar Timing Array and towards precision pulsar timing with SKA-mid"
Publications of the Astronomical Society of Australia

17. Middleton, et al., **(inc. Miles, M. T.)**, 2025, "A simple optimization for the MeerKAT pulsar timing array"
Monthly Notices of the Royal Astronomical Society

16. Grunthal, K. & Nathan, R., Thrane, E., Champion, D. J., **Miles, M. T.**, et al., 2025, "The MeerKAT Pulsar Timing Array: Maps of the gravitational-wave sky with the 4.5 year data release"
Monthly Notices of the Royal Astronomical Society

15. Shamohammadi, et al., **(inc. Miles, M. T.)**, 2025, "MeerKAT Pulsar Timing Array parallaxes and proper motions"
Monthly Notices of the Royal Astronomical Society

14. Kulkarni, A. D., Shannon, R. M., Reardon, D. J., **Miles, M. T.**, Bailes, M., Shamohammadi, M., 2023, "An insight into chromatic behaviour of jitter in pulsars and its modelling: A case study of

PSR J0437−4715"
Monthly Notices of the Royal Astronomical Society

13. Gitika, et al., **(inc. Miles, M. T.)**, 2023, "Flux density monitoring of 89 millisecond pulsars with MeerKAT"
Monthly Notices of the Royal Astronomical Society

12. Agazie, et al., **(inc. Miles, M. T.)**, 2023, "Comparing recent PTA results on the nanohertz stochastic gravitational wave background"
The Astrophysical Journal Letters

11. Zic, et al., **(inc. Miles, M. T.)**, 2023, "The Parkes Pulsar Timing Array Third Data Release"
Publications of the Astronomical Society of Australia

10. Reardon, et al., **(inc. Miles, M. T.)**, 2023, "The gravitational-wave background null hypothesis: Characterizing noise in millisecond pulsar arrival times with the Parkes Pulsar Timing Array"
The Astrophysical Journal Letters

9. Reardon, et al., **(inc. Miles, M. T.)**, 2023, "Search for an isotropic gravitational-wave background with the Parkes Pulsar Timing Array"
The Astrophysical Journal Letters

8. Nathan, R., **Miles, M. T.**, Ashton, G., Lasky, P. D., Thrane, E., Reardon, D. J., Shannon, R. M., Cameron, A. D., 2023, "Improving pulsar-timing solutions through dynamic pulse fitting"
Monthly Notices of the Royal Astronomical Society

7. Shamohammadi, et al., **(inc. Miles, M. T.)**, 2022, "Searches for Shapiro delay in seven binary pulsars using the MeerKAT telescope"
Monthly Notices of the Royal Astronomical Society

6. Curylo, et al., **(inc. Miles, M. T.)**, 2022, "Wideband timing of the Parkes Pulsar Timing Array UWL data"
American Astronomical Society

5. Spiewak, Renee, Bailes, Matthew, **Miles, M. T.**, et al., 2022, "The MeerTime Pulsar Timing Array–A Census of Emission Properties and Timing Potential"
Publications of the Astronomical Society of Australia

4. Goncharov, et al., **(inc. Miles, M. T.)**, 2021, "On the Evidence for a Common-spectrum Process in the Search for the Nanohertz Gravitational-wave Background with the Parkes Pulsar Timing Array"
The Astrophysical Journal Letters

3. Ackley, K, et al., **(inc. Miles, M. T.)**, 2020, "Neutron Star Extreme Matter Observatory: A kilohertz-band gravitational-wave detector in the global network."
Publications of the Astronomical Society of Australia

2. Norfolk, Brodie J., Casey, Andrew R., Karakas, Amanda I., **Miles, Matthew T.**, Kemp, Alex J., Schlaufman, Kevin C., Ness, Melissa, Ho, Anna Y. Q., Lattanzio, John C. and Ji, Alexander P., 2019, "Discovery of s-process enhanced stars in the LAMOST survey."
Monthly Notices of the Royal Astronomical Society

1. Kemp, Alex J., Casey, Andrew R., **Miles, Matthew T.**, Norfolk, Brodie J., Lattanzio, John C., Karakas, Amanda I., Schlaufman, Kevin C., Ho, Anna Y. Q., Tout, Christopher A., Ness, Melissa and Ji, Alexander P., 2018, "On the discovery of K-enhanced and possibly Mg-depleted stars throughout the Milky Way."
Monthly Notices of the Royal Astronomical Society

CONTRIBUTED / INVITED TALKS (NOT EXHAUSTIVE)

Lorentz Workshop *19/05/2026*
Propsects for accurate pulsar distance measurements

CSIRO ATNF seminar series *10/03/2026*
A new way to time pulsars

Benasque orbital resonance workshop *15/12/2025*
How pulsar timing arrays operate

Seminar series, Yale University *17/10/2025*
The future of gravitational wave detection with pulsar timing arrays

Amaldi Meeting *17/07/2025*
Forecasts for the future of the MeerKAT Pulsar Timing Array

Astronomical Society of Australia *11/07/2025*
Charlene Heisler prize talk

International Pulsar Timing Array (IPTA), annual conference *23/06/2025*
Updates from The MeerKAT Pulsar Timing Array

APS Meeting *17/03/2025*
The MeerKAT Pulsar Timing Array: Results of the first gravitational wave searches

DSA-2000, Working Group *09/01/2025*
The MeerKAT Pulsar Timing Array: Updates and recent results

International Astronomical Union (IAU), General Assembly *06/08/2024*
The MeerKAT Pulsar Timing Array: Results of the first gravitational wave searches

OzGrav Telecon *19/07/2024*
The MeerKAT Pulsar Timing Array: Results of the first gravitational wave searches

Seminar series, The University of Melbourne *10/07/2024*
What is a PTA, why you should be excited, and the first results of the MPTA

International Pulsar Timing Array (IPTA), annual conference *24/06/2024*
The MeerKAT Pulsar Timing Array: Results of the first gravitational wave searches

European Pulsar Timing Array (EPTA), Summer Conference *27/03/2024*
The MeerKAT Pulsar Timing Array: Results of the first gravitational wave searches

MeerKAT @5, Stellenbosch *20/02/2024*
The MeerKAT Pulsar Timing Array: Results of the first gravitational wave searches

Honours Colloquium Series, University of Cape Town *03/10/2023*
Pulsar Timing Arrays: What are they and why should you be excited

Amaldi 15, Premier International Conference on Gravitational Waves *18/07/2023*
Noise modelling and updated results from the MeerKAT Pulsar Timing Array

Astronomical Society of Australia, annual science meeting *30/06/2023*
The MeerKAT Pulsar Timing Array

International Pulsar Timing Array (IPTA), annual conference *21/06/2023*
Noise modelling and updated results from the MeerKAT Pulsar Timing Array

Department of Physics and Astrophysics seminar *12/05/2023*
A brief history of Gravitational Wave science

OzGrav Telecon *03/02/2023*
MeerKAT Pulsar Timing Array: A search for common signals

OzGrav Annual Retreat *24/11/2022*
The future of nHz-frequency gravitational wave astrophysics in the post-detection era

ACAMAR 8 *13/10/2022*
MeerKAT Pulsar Timing Array: First Data Release & updates

European Pulsar Timing Array (EPTA), Summer conference *21/09/2022*
MeerKAT Pulsar Timing Array: First Data Release & updates

Australasion Pulsar Meeting *09/08/2022*
MeerKAT Pulsar Timing Array: First Data Release & updates

NANOGrav VIPER group, Vanderbilt University, Nashville TN *14/07/2022*
MeerKAT Pulsar Timing Array overview

International Pulsar Timing Array (IPTA), annual conference *15/06/2022*
MeerTime MSP timing results

European Pulsar Timing Array (EPTA), annual conference *18/03/2022*
MeerTime MSP timing results

OzGrav Retreat *24/11/2021*
Techniques to constrain the emission of mode-changing millisecond pulsars

COLLABORATIONS

Member, Gamma-ray Pulsar Timing Array (GPTA) *Aug 2026 - Current*
Vanderbilt University

Associate Member, North American Nanohertz Observatory for Gravitational Waves (NANOGrav) *Jan 2025 - Current*
Vanderbilt University

Leader, MeerKAT Pulsar Timing Array (MPTA) *Mar 2020 - Current*
Vanderbilt University

Member, The ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav) *Mar 2020 - Current*
Swinburne University of Technology

Member, Parkes Pulsar Timing Array (PPTA) *Mar 2020 - Current*
Vanderbilt University

Member, International Pulsar Timing Array (IPTA) *Mar 2020 - Current*
Vanderbilt University
Co-Chair of the Gravitational Wave Analysis research group

AWARDS & NOTABLE STATISTICS

- 2025 Charlene Heisler prize: Awarded for the most outstanding PhD thesis in astronomy or a closely related field, accepted by an Australian University.

- Per Pimbblet et al., 2011, as a PhD student I ranked in the 99th percentile of Australian astronomers per the h-index citation statistic.
- 2nd and 4th place at the Australian University Rocketry Competition, for the 30,000 ft and 10,000 ft categories, respectively. (2019)
- Winning design for the International Medical Express UAV Challenge. (2018)

UNDERGRADUATE PROJECTS (SELECTED, NOT EXHAUSTIVE)

Final Year Project (Honours) - Aerospace Engineering Mar 2019 - Nov 2019
Monash Faculty of Engineering

Machine Learning and the Future of Australia's Energy Requirements Jun 2019
Monash University

Astrophysics Research Project Dec 2017 - Feb 2018
Monash School of Physics & Astronomy

High Powered Rocketry (HPR) - Propulsion Engineer Sept 2018 - Dec 2019
Monash Faculty of Engineering

EMPLOYMENT HISTORY (SELECTED, NOT EXHAUSTIVE)

Postdoctoral Research Fellow Jan 2025 - Current
Vanderbilt University

Postdoctoral Research Fellow Sept 2023 - December 2024
Swinburne University of Technology / OzGrav

Data Scientist Feb 2021 - July 2022
Eloque

Associate Consultant Jun 2018 - Feb 2020
JVAT Consulting

VOLUNTEERING & OUTREACH (SELECTED, NOT EXHAUSTIVE)

AstroTour Coordinator Aug 2022 - Sept 2023
Swinburne University of Technology

President Oct 2017 - Oct 2018
Society of Arts Students, Monash University